

NIKHILNANDALA

Data Engineer

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SUMMARY

Results-driven Data Engineer with around 5 years of experience designing, building, and optimizing large-scale data pipelines, cloud architectures, and analytics platforms for banking, fintech, e-commerce, logistics, and consulting domains. Proven expertise in **ETL development**, **big data processing** (Hadoop, Spark, Hive), **cloud data engineering** (AWS, GCP, Azure), and **data warehousing** (Snowflake, Amazon Redshift, Google BigQuery). Adept at **SQL optimization**, **dimensional modeling**, and **data quality governance**, delivering measurable gains in processing speed, cost efficiency, and SLA compliance. Skilled in **automation** with Apache Airflow, AWS Glue, and CI/CD pipelines, enabling near-real-time analytics for datasets exceeding billions of records daily. Strong track record of collaborating with cross-functional teams to deliver **business-critical insights**, reduce operational costs, and support enterprise-scale decision-making.

EXPERIENCE

Data Engineer | JP Morgan Chase & Co. | USA

February 2025 – Current

- **Engineered** a high-performance **PySpark** ETL workflow on Hadoop to process **2.5 billion+ daily transaction records**, reducing data processing time by **43%** and enabling near-real-time fraud detection for retail banking operations.
- **Optimized** complex **SQL** queries in Amazon Redshift for investment portfolio analytics, cutting query runtime from **12 minutes to under 3 minutes** and improving reporting SLA compliance from **93% to 99.5%**.
- **Architected** scalable **AWS Glue** pipelines to integrate **12 disparate treasury data sources** into a centralized data lake, increasing available curated datasets from **25 to 40** for downstream analytics teams.
- **Developed** robust **Snowflake** dimensional models for wealth management KPIs, improving metric consistency across **5 business units** and reducing manual reconciliation effort by **18 hours per week**.
- **Deployed** automated data quality checks in **Apache Airflow** for commercial banking loan datasets, preventing ingestion of **over 4 million erroneous records annually** before they reached BI dashboards.
- **Created** interactive **Power BI** dashboards for cross-asset risk exposure, enabling traders and risk officers to monitor **20 million+ daily trade records** and reduce risk assessment cycles from **48 hours to under 12 hours**.

Data Engineer | Uber | USA

March 2024 – February 2025

- **Engineered** a high-throughput data ingestion framework using **PySpark** on **Hadoop** to process 3.1 TB/day of trip, payment, and driver telemetry data, enabling near-real-time analytics and cutting daily batch runtime by 6.5 hours.
- **Optimized** complex **SQL** queries and partitioning strategies in **Amazon Redshift** for marketplace demand forecasting models, reducing average query execution time from 18 minutes to under 7 minutes and lowering compute costs by \$79,000 annually.
- **Automated** 215 daily ETL workflows with **Apache Airflow** for Uber Eats order, menu, and delivery datasets, achieving 99.8% ontime execution and freeing up 780+ engineering hours per year previously spent on manual job reruns.
- **Migrated** 43 TB of historical ride-sharing data from **on-premise PostgreSQL** to **Google BigQuery**, enabling analysts to run ad-hoc revenue and demand queries in under 30 seconds versus 2–3 minutes previously.
- **Implemented** robust data quality checks using **AWS Glue** with schema validation and anomaly detection, blocking ingestion of ~1.26 million corrupt or incomplete records each month and safeguarding compliance-critical datasets.
- **Developed** dimensional models (Star/Snowflake schema) for **Tableau** dashboards tracking driver incentives across 150+ U.S. cities, unlocking insights that guided \$12.4M in optimized incentive spending over a fiscal year.
- **Deployed** infrastructure-as-code with **Terraform** for provisioning scalable **AWS EMR** clusters, cutting environment setup time from 3 days to 3.5 hours and supporting surge loads of up to 95,000 concurrent trip events without downtime.

Data Engineer | Capgemini | India

June 2021 – August 2022

- **Engineered** a scalable ETL framework using **Apache Airflow**, automating ingestion from 12+ disparate sources for a global retail client, cutting manual workload by 320 hours per month and enabling near real-time reporting for 45 business units.
- **Optimized** complex analytical queries in **Hive** by applying partitioning and bucketing strategies, reducing average execution time from 18 minutes to under 4 minutes across 75 million records, which accelerated churn analysis cycles by 3 days.
- **Deployed** a secure, serverless data processing solution on **AWS Glue** for a financial services project, processing over **2 TB** of daily transactions while maintaining 99.99% job success rate and meeting PCI-DSS compliance audits.
- **Modeled** enterprise data using **Snowflake's** star schema for a healthcare provider, supporting analytics on 28 million patient records and improving dashboard load times from 15 seconds to under 5 seconds.
- **Automated** CI/CD pipelines with **Jenkins**, enabling continuous deployment of PySpark jobs to EMR clusters, reducing release cycles from 14 days to 1 day and saving an estimated ₹18 lakh annually in operational costs.
- **Integrated** structured and semi-structured IoT data into **Cassandra**, enabling real-time analytics for 1.2 million sensor readings per hour and increasing predictive maintenance accuracy by 17% for manufacturing equipment.

- **Delivered** executive dashboards in **Power BI** for a logistics company, consolidating KPIs from 32 warehouses and cutting monthly reporting effort from 5 days to 4 hours, directly improving decision turnaround for supply chain planning.

Data Engineer | KPMG | India

July 2019 – May 2021

- **Engineered** a scalable data ingestion framework in **Python**, consolidating structured and semi-structured feeds from 12+ client systems, reducing onboarding time for new datasets from 10 days to **6 days**.
- **Optimized** ETL workflows on **Apache Spark** (PySpark) to process **1.8 billion audit records** monthly, cutting average runtime from 7 hours to 2.5 hours and saving 420 compute hours per month.
- **Orchestrated** cloud-based data pipelines on **AWS Glue** for risk analytics, enabling near real-time updates of **2 TB/day** of regulatory data across 3 global audit teams.
- **Deployed** a centralized warehouse in **Snowflake**, storing and serving **15 TB** of cross-domain financial data with query performance improved from 90 seconds to 18 seconds.
- **Designed** optimized analytical queries in **PostgreSQL**, reducing report generation time from 12 minutes to under 5 minutes for compliance audits impacting **50+ government departments**.
- **Automated** workflow scheduling in **Apache Airflow**, integrating 54 dependent jobs, achieving **99.9% pipeline uptime** over 12 consecutive months.
- **Implemented** a CI/CD process with **Jenkins**, delivering 180+ production deployments annually with zero downtime and reducing release turnaround from 7 days to under 24 hours.
- **Modeled** a dimensional data structure using **Star Schema**, ensuring validated lineage for **20+ KPIs** used in quarterly regulatory filings worth ₹8,500 crore in reported assets.
- **Delivered** interactive risk dashboards in **Tableau**, consolidating 5 disparate data sources and helping audit managers cut decisionmaking time from 2 days to **less than 8 hours**.

TECHNICALSKILLS

- **Programming & Scripting:** Python, SQL, Scala, Shell Scripting
- **Big Data Technologies:** Hadoop, Spark (PySpark/Scala), Hive, Pig, HDFS
- **Cloud Platforms:** AWS (S3, EMR, Lambda, Redshift, Glue), Azure Data Factory, GCP (BigQuery, Cloud Storage)
- **ETL & Data Pipelines:** Apache Airflow, Talend, Informatica, NiFi, AWS Glue
- **Data Warehousing:** Amazon Redshift, Snowflake, Google BigQuery, Teradata
- **Databases:** PostgreSQL, MySQL, Oracle, NoSQL (MongoDB, Cassandra)
- **DevOps & CI/CD:** Git, Jenkins, Docker, Terraform
- **Data Modeling & Governance:** Dimensional Modeling, Star/Snowflake Schema, Data Lineage, Data Quality
- **Analytics & BI Tools:** Tableau, Power BI, Looker, Superset
- **Version Control & Collaboration:** GitHub, Bitbucket, Jira, Confluence

EDUCATION

Master of Science in Information Technology and Management

Webster University | Missouri, USA

August 2022 – May 2024

Bachelor of Technology in Computer Science

Jawaharlal Nehru Technological University | Hyderabad, India

July 2015 – December2019